

KEY FINDING — The Laterite Inversion: best material at asset scale (1 km) becomes worst at network scale (100 km) — a scale-dependent trade-off invisible to single-scale LCA

20-year horizon · 7 policy scenarios · 100 km network · 2,000 Monte Carlo runs · 9,216 Sobol samples · 10,000 MCDA draws

1 Introduction & Background

Nigeria relies on roads for ~80% of freight movement, yet over 60% of the paved network is in poor condition. Routine over-axle loading — reaching 200% of legal limits — accelerates deterioration far beyond temperate benchmarks [1].

Three materials compete: conventional asphalt, sealed laterite (locally available iron-rich gravel), and plastic-modified asphalt (PMA). No prior study couples asset-level LCA to network-level system dynamics — leaving scale-dependent trade-offs hidden from policy-makers.

	Asset scale (1 km)	Network scale (100 km)
Multi-metric (cost + CO ₂ + condition)	Egilmez & Tatari (2012)	Ruiz & Guevara (2020) Shiboub & Assaf (2022) ★ THIS STUDY Digital-twin coupling
Single-metric (condition only)	Talaghat et al. (2024)	Linard (2009, 2010)

Fig. 1: Research positioning. The top-right cell (network-scale, multi-metric) was previously unoccupied. This study fills it.

2 Research Questions

- RQ1** What are the 20-year life-cycle cost, CO₂ and condition trajectories for each material at asset level?
- RQ2** How does one-way coupling of asset outputs into a system-dynamics model enable network-level interpretation?
- RQ3** How do budget, adoption mix, and axle-load enforcement affect network backlog, cost, and emissions?

Answers to Research Questions

- A1** Laterite: lowest 20-yr NPV (£79,688) and CO₂ (236.7 t). PMA: longest service life (Yr 12 first overlay). Asset-level cost and carbon rankings are robust across all 2,000 Monte Carlo draws.
- A2** Coupling reveals The Laterite Inversion: asset-optimal material produces the worst network backlog. This scale-dependent trade-off cannot be detected by single-scale asset LCA.
- A3** Budget is non-binding above £346k/yr (S0 = S1 = S2 identically). WIM axle-load enforcement is the only effective lever, reducing Yr-20 backlog from 17.05 → 10.68 km (~37%).

3 Methodology Pipeline

1	CALIBRATE	PIARC tropical curves [1] + literature
2	ASSET MC	MATLAB · 3 materials · N = 2,000
3	UNIT OUTPUTS	TTL · £/km · t CO ₂ /km → feed network
4	NETWORK SD	Vensim · 100 km · 7 scenarios · 20 yr [3]
5	SOBOL	Variance decomp. · N = 9,216 (Saltelli)
6	MCDA	10,000 Dirichlet weight draws → rank robustness

4 Network Scenario Design (7 scenarios)

S0	Baseline	100% asphalt; calibrated budget; overload multiplier ODM = 1.0
S1	Low-budget ×0.7	70% of baseline annual road agency budget
S2	High-budget ×1.3	130% of baseline annual road agency budget
S3	Plastic-push	40% PMA / 40% asphalt / 20% laterite rehabilitation mix
S4	Laterite-heavy	50% laterite / 30% asphalt / 20% PMA rehabilitation mix
S5	Strong enforcement	WIM enforcement reduces ODM: 1.0 → 0.6 (~40%)
S6	Combined policy ★	S5 enforcement + S3 plastic mix — best on all 4 metrics

References

- [1] PIARC (2022). Pavement Management Systems. World Road Association. [2] Torres Martinez et al. (2018). WIM cost-benefit analysis. Road Mater. Pavement Des. [3] Ruiz & Guevara (2020a,b). SD for road networks. Transp. Res. D. [4] Talaghat et al. (2024). Pavement LCA review. J. Clean. Prod. [5] Shiboub & Assaf (2022). Network LCCA. Int. J. Pave. Res. Technol. [6] Linard (2009, 2010). SD in transport infrastructure. Syst. Dyn. Rev.



Scan to access code, Vensim model, data & full dissertation on GitHub

5 Asset-Level Results (1 km, 20-year baseline)

Material	NPV Cost	Const. CO ₂	W-L CO ₂	1st Overlay
Conventional asphalt	£99,872	195.9 t	335.4 t	Year 10
Sealed laterite	£79,688 ✓	44.7 t ✓	236.7 t ✓	Year 8
Plastic-modified (PMA)	£110,692	132.1 t	253.9 t	Year 12

Asphalt construction CO₂ (195.9 t) = 58% of whole-life total — contradicting temperate-climate use-phase dominance. Tropical overloading creates a construction-dominant emissions profile absent from OECD LCA literature.

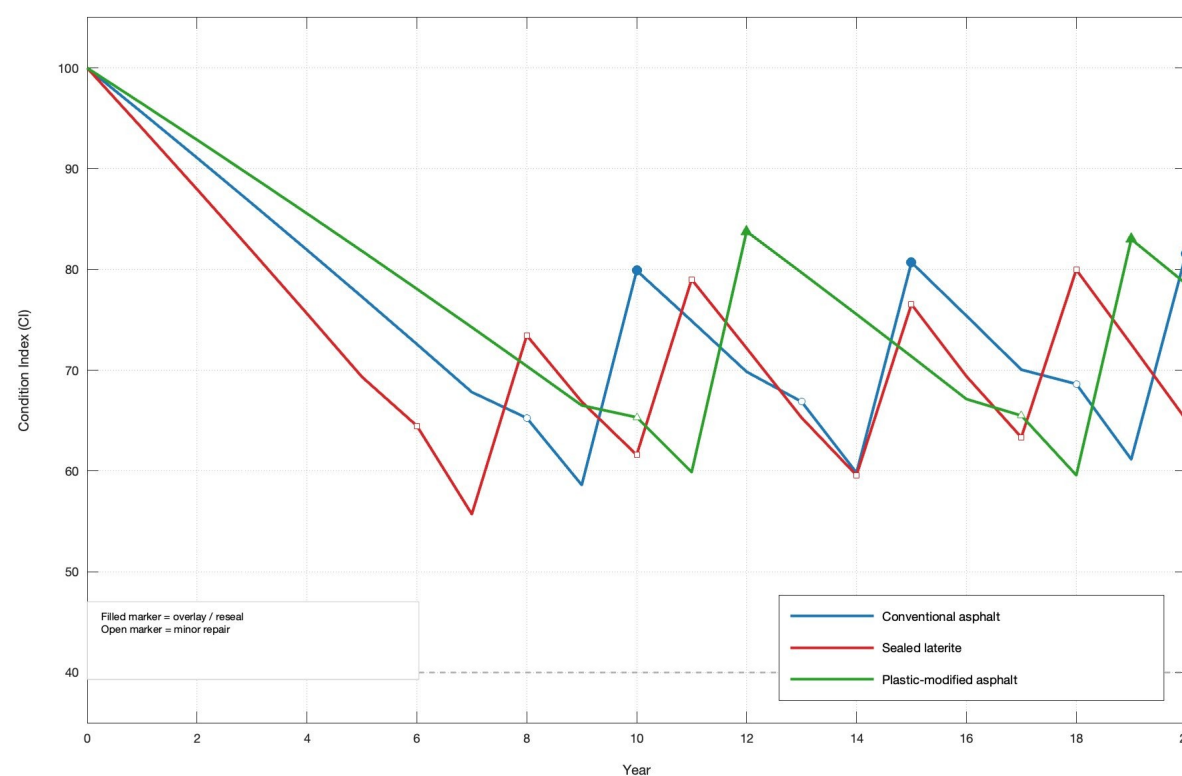


Fig. 2: CI vs time. The line staying highest longest is PMA (most durable). The line dropping first is laterite (Yr 8 overlay). Filled markers = overlay events.

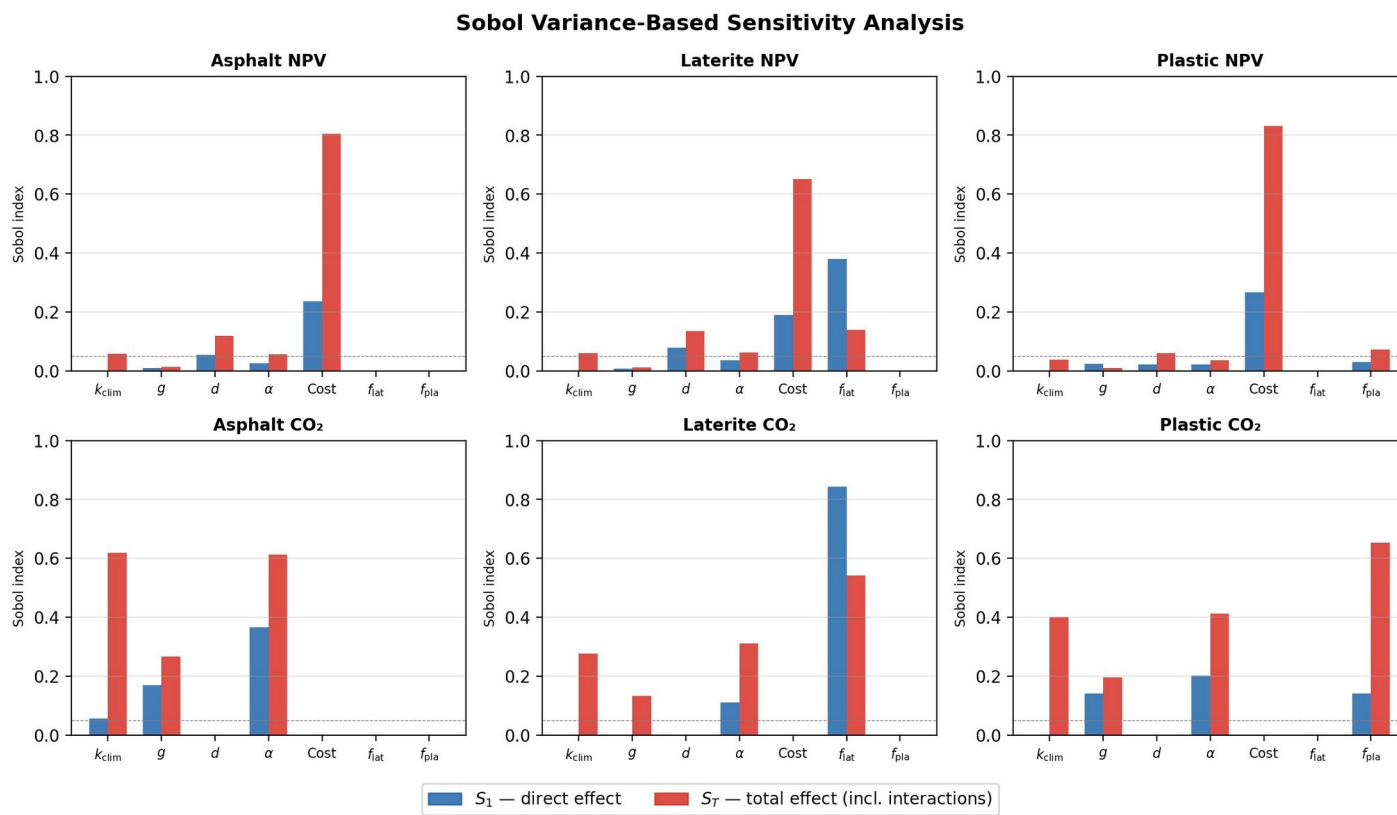
6 Sensitivity & Uncertainty Analysis

Monte Carlo (N = 2,000):

CO₂ ranking is stochastically dominant — laterite 95th-pct (88.9 t) < plastic 5th-pct (109.0 t). NPV rankings overlap: laterite median (£89,874) lowest, 95th-pct (£135,945) overlaps other materials.

Sobol decomposition (N = 9,216):

NPV driven by unit cost factor (S_T = 0.65–0.83). Asphalt CO₂: k_{climate} S₁ = 0.06 but S_T = 0.62 — 91% of variance from interaction. OAT methods systematically understate CO₂ uncertainty.



Monte Carlo Distributions (N = 2,000)

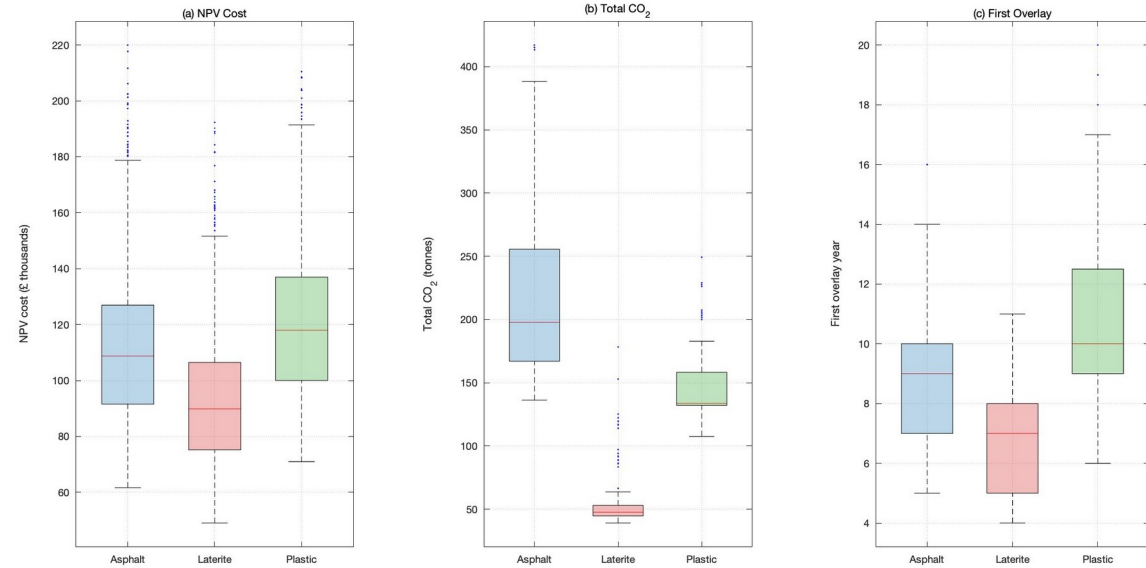


Fig. 4: MC distributions. Laterite CO₂ box sits entirely below the others — stochastic dominance, robust to all parameter draws.

Key Metrics at a Glance

1.4–3.8
yrWIM payback
(agency savings only)£88,481
/yrBudget-equivalent
benefit of enforcement

100%

S6 ranks 1st in
all 10,000 MCDA draws

Validation & Limitations

✓ Validation

- PIARC (2022) tropical deterioration curves
- SD structure: Ruiz & Guevara (2020)
- Sobol convergence at 5% tolerance
- LCA boundary aligned with EN 15804

! Limitations

- Tropical analogue calibration (not Nigerian)
- One-way coupling only (no feedback)
- Current climate (no IPCC trajectories)
- Homogeneous network (no spatial variation)

7 Network-Level Results (100 km, 7 scenarios)

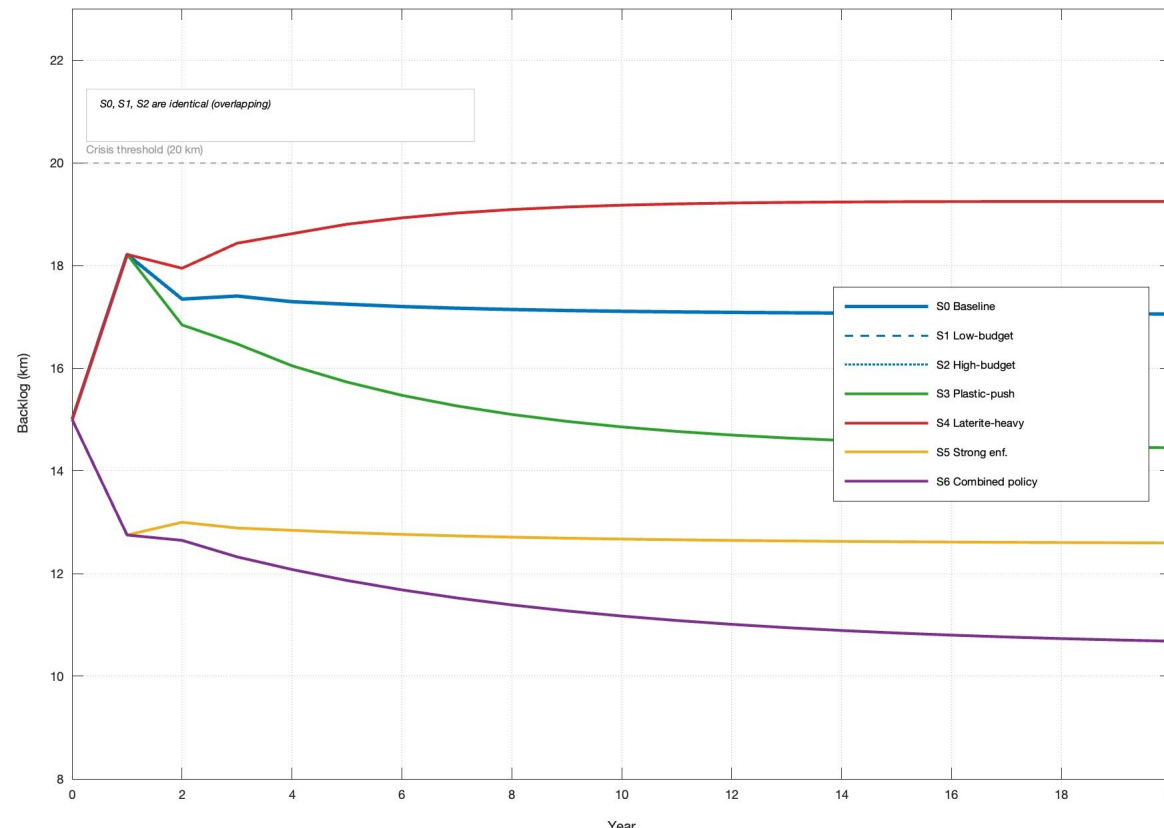


Fig. 5: Network backlog over 20 yr. The top red line is S4 at 19.24 km (Inversion). Three lines overlap in the middle at 17.05 km — those are S0, S1 and S2 (budget non-binding). The bottom purple line is S6 at 10.68 km (combined policy).

Scenario	Yr-20 Backlog	Cum. Cost	W-L CO ₂
S0 Baseline	17.05 km	£6.79M	17,929 t
S1 Low-budget	17.05 km	£6.79M	17,929 t
S2 High-budget	17.05 km	£6.79M	17,929 t
S3 Plastic-push	14.45 km	£6.94M	17,179 t
S4 Laterite-heavy	19.24 km	£6.04M	16,507 t
S5 Enforcement	12.60 km	£5.09M	16,855 t
S6 Combined ★	10.68 km	£5.26M	16,182 t

Why WIM enforcement is so powerful: pavement damage $\propto$ load⁴ (AASHO Road Test, 1962). A truck at 2× legal load does 16× the damage of a legal one [2].

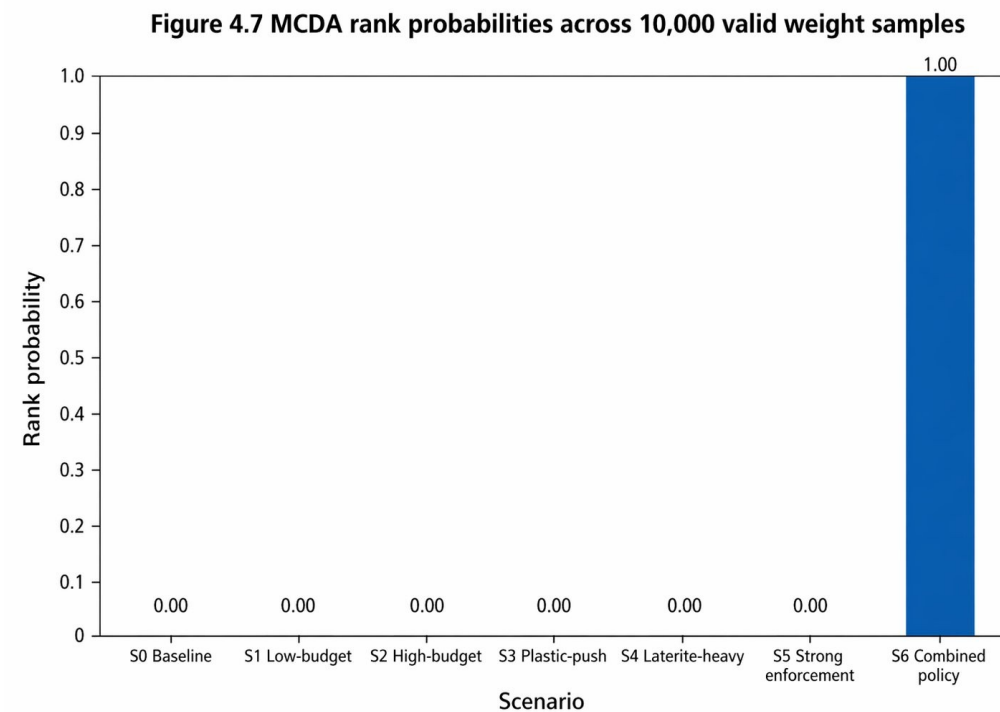


Fig. 6: The single bar reaching 1.0 is S6 — it ranks first in every one of 10,000 Dirichlet weight samples. Other scenarios never rank first under any weighting.

8 Discussion & Conclusions

① The Laterite Inversion — Asset-Optimal ≠ Network-Optimal

Asset Scale (1 km) ✓	Network Scale (100 km) ✗
NPV: £79,688 ✓	Yr-20 backlog: 19.24 km ✗
WL CO ₂ : 236.7 t ✓	vs S0: +13% ✗
TTL: 8 yr	MCDA rank: 5th / 7 ✗
MC CO ₂ : 1st / 3 ✓	cum. cost: £6.04M
BEST	WORST

TTL = 3.33 yr
(network-scale variance)

overwhelms
rehab queue

Hidden from conventional single-scale LCA — requires digital-twin coupling to detect

② Budget Non-Binding

Binding constraint on Nigerian network performance is deterioration rate, not budget. S0 = S1 = S2 identically. Policy should redirect resources to enforcement, not more rehabilitation contracts.

③ Multiplicative CO₂ Uncertainty

Climate and overload factors interact multiplicatively. OAT methods miss 91% of asphalt CO₂ variance. Sobol decomposition is necessary for carbon-critical infrastructure assessment in tropical contexts.

9 Policy Recommendations

Immediate Install WIM axle-load enforcement. Capital cost £120k–320k; payback 1.4–3.8 yr from agency savings alone [2]. Budget-equivalent benefit: £88,481/yr per 100 km.

Conditional Expand laterite only where corridor TTL evidence ≥ 5 yr. Survey justified at P(TTL≥5yr) = 40% (EVPI = £78/km). Wrong-decision cost: £437/km.

Strategic Adopt PMA where CO₂ reduction has policy value. Advantage over asphalt doubles by Year 25 and is robust to all Monte Carlo draws.

10 Future Work

- In-situ TTL measurement (FERMA partnership) to resolve laterite inversion (EVPI = £78/km)
- Regional disaggregation: PMA in humid south, polymer-asphalt in arid north, laterite for rural
- Two-way coupling (condition → user costs → GDP) + IPCC climate trajectories

RECOMMENDED ACTION — S6 Combined Policy: WIM axle-load enforcement + 40% PMA / 40% asphalt / 20% laterite mix. –37% Yr-20 backlog · payback 1.4–3.8 yr · ranks 1st in 100% of MCDA draws